

Standard Operating Procedure for Ubuntu Server Provisioning, Network Configuration, and Hardening

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1. INTRODUCTION

1.1 Purpose of the Manual

This Standard Operating Procedure (SOP) provides explicit, reproducible technical steps to deploy, configure, verify, and secure an enterprise **Ubuntu Server** instance. It is engineered so that any system administrator or network operations engineer can replicate this deployment without additional verbal instructions.

1.2 Scope and Target Audience

This guide covers provisioning on bare metal or hypervisor platforms (such as Oracle VM VirtualBox), network stack verification, initial security patching, and base diagnostics.

1.3 System Prerequisites & Baseline Specifications

- **Hypervisor Platform:** Oracle VM VirtualBox v7.0 or above
- **OS Media:** Ubuntu Server 64-bit ISO
- **vCPU Allocation:** 1 to 2 Virtual Cores
- **Memory (RAM):** 2048 MB (2 GB) Minimum
- **Virtual Storage:** 20.0 GB Dynamically Allocated Disk (Standard ext4 filesystem)
- **Network Interface:** Attached to NAT or Bridged Adapter

2. UBUNTU SERVER INSTALLATION GUIDE

2.1 Virtual Machine Initialization

1. Launch **Oracle VirtualBox** and click **New**.
2. Set Name: Ubuntu-Server-Week03, Type: Linux, Version: Ubuntu (64-bit).
3. Allocate 2048 MB Base Memory and assign at least 1 vCPU.
4. Create a virtual hard disk with a minimum capacity of 20.00 GB.
5. Under **Settings \$rightarrow\$ Storage**, attach the Ubuntu Server ISO into the Optical Drive.
6. Under **Settings \$rightarrow\$ Network**, set **Adapter 1** to NAT (or Bridged Adapter depending on corporate lab topology).

2.2 Storage Partitioning & OS Deployment

1. Start the Virtual Machine and select **Try or Install Ubuntu Server**.
2. Select installation language: **English** and confirm the keyboard layout.
3. Configure the Network Connection: Ensure the default Ethernet interface (enp0s3) obtains an automatic DHCP address.
4. Storage Configuration: Select **Use an entire disk**, confirm the virtual drive assignment, and select **Done** to write changes to disk.

2.3 User Account Creation & Base Credentialing

1. Set the profile setup fields:
 - **Your Name:** Justine Mark
 - **Your Server's Name:** server01
 - **Pick a Username:** gahi
 - **Password:** Choose an enterprise-grade passphrase conforming to complexity standards.
2. Select **Install OpenSSH Server** for secure remote CLI management.
3. Complete the installation and reboot the virtual machine. Eject the virtual installation media when prompted.

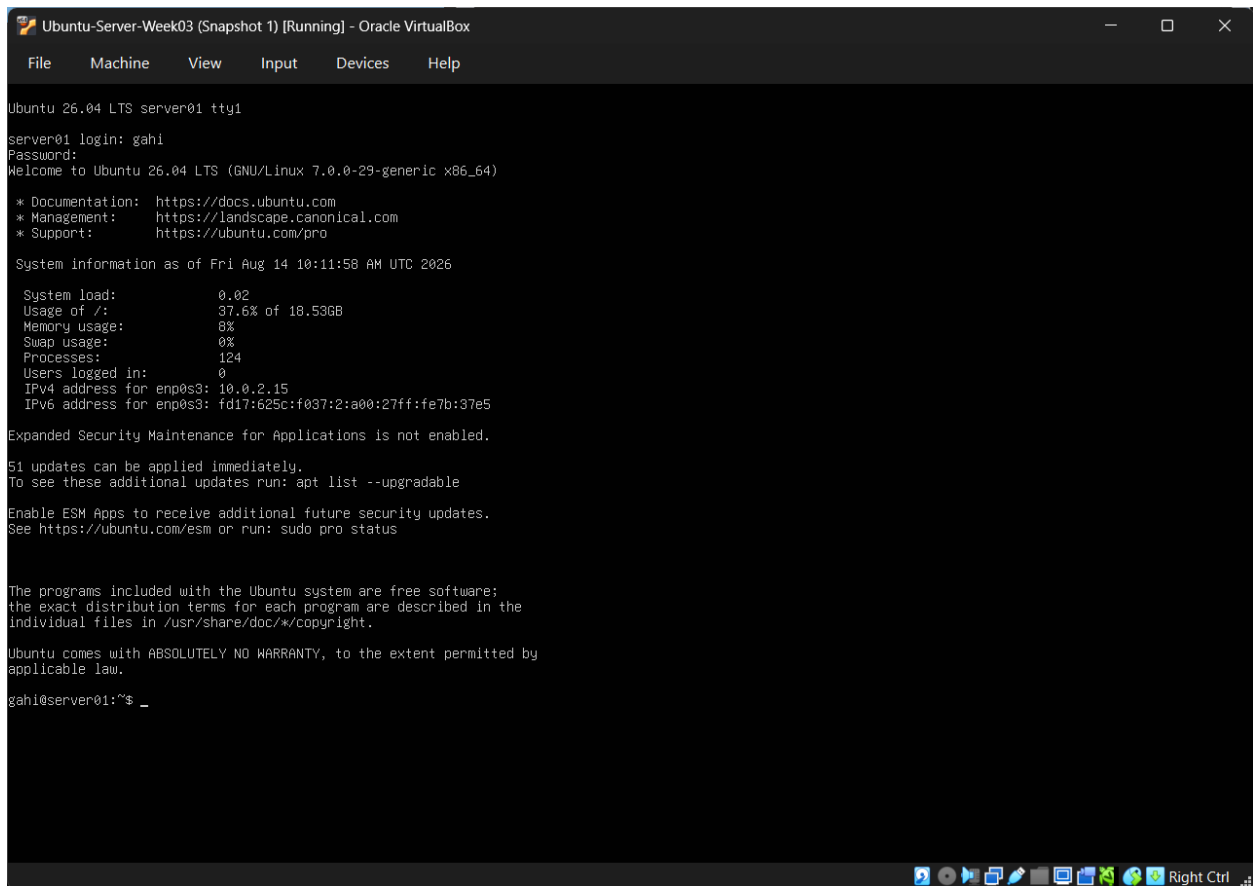
3. INITIAL SERVER CONFIGURATION

3.1 First Login & Host Authentication

Upon completion of the system boot sequence, log in using the provisioned credentials through the virtual console (tty1).

Ubuntu 26.04 LTS server01 tty1 server01 login: gahi Password: [ENTERED]

Verification Point: Successful authentication will display the message of the day (MOTD) with memory usage, system load, and storage metrics.



```
Ubuntu-Server-Week03 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS server01 tty1
server01 login: gahi
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 14 10:11:58 AM UTC 2026
System load:          0.02
Usage of /:           37.6% of 10.53GB
Memory usage:         8%
Swap usage:           0%
Processes:            124
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe7b:37e5

Expanded Security Maintenance for Applications is not enabled.

51 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

gahi@server01:~$ _
```

3.2 Hostname Configuration & Identification

Verify that the hostname is set accurately to identify this node across the local enterprise network.

hostname

- **Expected Output:** server01

```

Ubuntu 26.04 LTS server01 tty1
server01 login: gahi
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 14 10:11:58 AM UTC 2026

System load:          0.02
Usage of /:           37.6% of 18.53GB
Memory usage:         8%
Swap usage:           0%
Processes:            124
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe7b:37e5

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Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

gahi@server01:~$ hostname
server01
gahi@server01:~$ hostname hostname

```

3.3 Network Adapter & IP Address Verification

Check the state of the network interfaces to confirm the system received an IP configuration from DHCP.

ip addr

Loopback (lo): 127.0.0.1/8 (Local internal address)

- **Primary Adapter (enp0s3):** 10.0.2.15/24 (State: UP, Broadcast: 10.0.2.255)

```

Usage: hostname [-b] {hostname|-F file}      set host name (from file)
hostname [-a|-A|-d|-f|-i|-I|-s|-y]         display formatted name
hostname                                     display host name

        {yp,nis,}domainname {nisdomain|-F file} set NIS domain name (from file)
        {yp,nis,}domainname                 display NIS domain name

        dnsdomainname                       display dns domain name

        hostname -V|--version|-h|--help     print info and exit

Program name:
        {yp,nis,}domainname=hostname -y
        dnsdomainname=hostname -d

Program options:
-a, --alias            alias names
-A, --all-fqdns        all long host names (FQDNs)
-b, --boot             set default hostname if none available
-d, --domain           DNS domain name
-f, --fqdn, --long     long host name (FQDN)
-F, --file             read host name or NIS domain name from given file
-i, --ip-address       addresses for the host name
-I, --all-ip-addresses all addresses for the host
-s, --short            short host name
-y, --yp, --nis        NIS/YP domain name

Description:
This command can get or set the host name or the NIS domain name. You can
also get the DNS domain or the FQDN (fully qualified domain name).
Unless you are using bind or NIS for host lookups you can change the
FQDN (Fully Qualified Domain Name) and the DNS domain name (which is
part of the FQDN) in the /etc/hosts file.
gahi@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:7b:37:e5 brd ff:ff:ff:ff:ff:ff
    altname enx0800277b37e5
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85624sec preferred_lft 85624sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7b:37e5/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86240sec preferred_lft 14240sec
    inet6 fe80::a00:27ff:fe7b:37e5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
gahi@server01:~$

```

4. SERVER VERIFICATION PROCEDURES

4.1 Internal Loopback & Interface Up-State Validation

Ensure the network layer responds properly by confirming the interface state:

- Execute `ip link show enp0s3` and verify the status flag reports `<BROADCAST,MULTICAST,UP,LOWER_UP>`.

4.2 Gateway & External DNS Reachability Testing

Confirm that the server can resolve external domain names via DNS and route ICMP traffic to the Internet:

`ping -c 4 google.com` Target Output: 4 packets transmitted, 4 packets received, 0% packet loss.

```

Program name:
  {yp,nis,}domainname=hostname -y
  dnsdomainname=hostname -d

Program options:
  -a, --alias          alias names
  -A, --all-fqdns      all long host names (FQDNs)
  -b, --boot           set default hostname if none available
  -d, --domain         DNS domain name
  -f, --fqdn, --long   long host name (FQDN)
  -F, --file           read host name or NIS domain name from given file
  -i, --ip-address     addresses for the host name
  -I, --all-ip-addresses all addresses for the host
  -s, --short          short host name
  -y, --yp, --nis      NIS/YP domain name

Description:
  This command can get or set the host name or the NIS domain name. You can
  also get the DNS domain or the FQDN (fully qualified domain name).
  Unless you are using bind or NIS for host lookups you can change the
  FQDN (Fully Qualified Domain Name) and the DNS domain name (which is
  part of the FQDN) in the /etc/hosts file.

gahi@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
   link/ether 08:00:27:7b:37:e5 brd ff:ff:ff:ff:ff:ff
   altname enx0800277b37e5
   inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 85624sec preferred_lft 85624sec
   inet6 fd17:625c:f037:2:a00:27ff:fe7b:37e5/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86240sec preferred_lft 14240sec
   inet6 fe80:a00:27ff:fe7b:37e5/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
gahi@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.238) 56(84) bytes of data.
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=1 ttl=64 time=11.2 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=2 ttl=64 time=11.2 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=3 ttl=64 time=8.28 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=4 ttl=64 time=10.3 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3045ms
rtt min/avg/max/mdev = 8.282/10.263/11.221/1.200 ms
gahi@server01:~$

```

4.3 Enterprise Repository Sync & Security Upgrades

Execute a system update to bring all security definitions, kernel binaries, and core libraries to the latest stable state:

```
sudo apt update && sudo apt upgrade -y
```

Critical Operation: Allow update-initramfs to automatically regenerate the initial RAM disk image (/boot/initrd.img-7.0.0-29-generic) to ensure uninterrupted system startups following kernel upgrades.


```

Setting up bind9-host (1:9.20.24-1ubuntu0.1)...
Setting up linux-firmware-misc (20260319.git217ca6e4-0ubuntu2.2)...
Setting up linux-firmware-qualcomm-graphics (20260319.git217ca6e4-0ubuntu1.1)...
Setting up fwupd (2.1.1-1ubuntu3.1)...
fwupd-refresh.service is a disabled or a static unit not running, not starting it.
fwupd.service is a disabled or a static unit not running, not starting it.
Setting up sg3-utils (1.48-3ubuntu3.1)...
Setting up cloud-init (26.1-0ubuntu3~26.04.1)...
Setting up ubuntu-minimal (1.570.2)...
Setting up sg3-utils-udev (1.48-3ubuntu3.1)...
update-initramfs: deferring update (trigger activated)
Setting up bind9-dnsutils (1:9.20.24-1ubuntu0.1)...
Setting up ubuntu-standard (1.570.2)...
Setting up apport-core-dump-handler (2.34.1-0ubuntu0.1)...
Setting up apport (2.34.1-0ubuntu0.1)...
apport-autoreport.service is a disabled or a static unit not running, not starting it.
Setting up ubuntu-server (1.570.2)...
Setting up ubuntu-server-minimal (1.570.2)...
Processing triggers for plymouth (24.004.60+git20250831.4a3c171d-0ubuntu0)...
Processing triggers for libc-bin (2.40-2ubuntu2.3)...
Processing triggers for rsyslog (8.2512.0-1ubuntu4.1)...
Processing triggers for systemd (259.5-0ubuntu3.4)...
Processing triggers for nan-db (2.13.1-1build1)...
Processing triggers for dbus (1.16.2-2ubuntu4)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
^[[    Processing triggers for procps (2:4.0.4-9ubuntu1)...
Processing triggers for install-info (7.2-5ubuntu2)...
Processing triggers for plymouth-theme-ubuntu-text (24.004.60+git20250831.4a3c171d-0ubuntu0)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service

No containers need to be restarted.

User sessions running outdated binaries:
gahi @ user manager: (sd-pam) [1467]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
gahi@server01:~$

```

5. TROUBLESHOOTING GUIDE

Issue Encountered	Root Cause	Resolution / Standard Procedure
Command-Line Syntax Help Triggered	Entering hostname hostname instead of running hostname alone or using hostnamectl.	Run hostname to query, or use sudo hostnamectl set-hostname <new-name> to modify.
Service Restarts Deferred (needrestart)	Daemons such as dbus.service or user manager sessions are held active during package upgrades.	Allow automated triggers to complete, or manually restart deferred services via sudo systemctl restart <service_name> after maintenance hours.

Issue Encountered	Root Cause	Resolution / Standard Procedure
No Internet Connectivity / DNS Failure	Virtual adapter is unplugged, misconfigured in VirtualBox, or DHCP lease failed.	Check VM Network settings, restart the link using <code>sudo ip link set enp0s3 down && sudo ip link set enp0s3 up</code> , or run <code>sudo netplan apply</code> .

6. BEST PRACTICES FOR SYSTEM ADMINISTRATORS

- **6.1 Routine Patch Management:** Establish a scheduled maintenance window for running `sudo apt update && sudo apt upgrade -y` to address zero-day vulnerabilities.
- **6.2 Principle of Least Privilege:** Never operate directly out of the root account. Always execute administrative actions through designated sudo privileges assigned to personal user accounts (e.g., gahi).
- **6.3 Hypervisor Snapshot Strategy:** Always generate a VirtualBox Snapshot (e.g., Snapshot 1 - Clean Base Install) immediately following successful configuration and verification before deploying production services or staging applications.
- **6.4 Host Hardening:** Ensure SSH root login is disabled in `/etc/ssh/sshd_config` (`PermitRootLogin no`) and configure an active firewall profile via `ufw`.

7. REFERENCES

- Canonical Ltd. (2024). *Official Ubuntu Server Guide & Deployment Documentation*. <https://docs.ubuntu.com>
- Oracle Corporation. (2023). *Oracle VM VirtualBox User Manual (Networking and Storage Configuration)*. <https://www.virtualbox.org/manual/>

- Linux Foundation. (2023). *Systemd System and Service Administration Standards*.
<https://systemd.io/>